VIT®

Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

NAME: THILIPAN

REGISTRATION NO: 23MID0272

ADVANCED PREDICTIVE ANALYSIS

EXECUTIVE SUMMARY:

This study developed and evaluated an interpretable machine learning pipeline for disease-status classification using the Breast Cancer Wisconsin (Diagnostic) dataset. The primary objective was to classify breast tumors as malignant or benign using Decision Tree classifiers while maintaining interpretability, minimizing overfitting, and ensuring reproducible evaluation.

The experimental pipeline progressed systematically from a Dummy Classifier baseline to an unconstrained Decision Tree, followed by a pre-pruned Decision Tree to reduce model complexity and improve generalization. Model performance was assessed using Accuracy, Balanced Accuracy, Sensitivity (Recall), Specificity, Precision, F1-Score, ROC-AUC, PR-AUC, and B rier Score.

The pre-pruned Decision Tree was selected as the final model because it achieved strong predictive performance while substantially reducing tree complexity. The final model achieved an Accuracy of 90.35%, Sensitivity of 80.95%, Specificity of 95.83%, ROC-AUC of 0.915, and PR-AUC of 0.878 on the held-out test set. Compared with the unconstrained tree, the pruned model produced fewer decision rules, improved interpretability, and reduced the likelihood of overfitting while maintaining competitive predictive performance.

STAKEHOLDER ANALYSIS

The primary stakeholders include:

- Healthcare researchers evaluating predictive models.
- Hospital analytics teams prioritizing patients for further diagnostic review.
- Medical AI researchers studying interpretable machine learning.
- Students learning responsible predictive analytics in healthcare.

PROJECT OBJECTIVE:

The primary objective of this project is to develop a reliable disease classification system capable of distinguishing between benign and malignant breast tumors using supervised machine learning techniques. The specific objectives include:

• Study the dataset characteristics.

• Perform data preprocessing.

• Build Decision Tree models.

• Optimize model performance.

• Compare Decision Tree with Random Forest.

• Evaluate prediction accuracy.

• Interpret model decisions.

OPERATIONAL DECISION CONTEXT

The objective is to develop an interpretable classification model capable of predicting whether a breast tumor is malignant or benign using numerical measurements extracted from digitized fine-needle aspirate images. The model is intended solely for educational and research purposes and is not a clinical diagnostic system.

COST OF MODEL ERRORS

False Negatives (Malignant classified as Benign)

- Delay diagnosis
- Delay treatment
- Potentially severe clinical consequences
- Highest priority error

False Positives (Benign classified as Malignant)

- Additional medical tests

- Increased healthcare cost
- Patient anxiety
- Lower clinical risk than false negatives

Because of these consequences, Sensitivity is considered one of the most important evaluation metrics.

SUCCESS THRESHOLD CRITERIA

The predictive pipeline should:

- Outperform a Dummy Classifier baseline.
- Maintain high Sensitivity while preserving good Specificity.
- Achieve strong ROC-AUC and PR-AUC values.
- Demonstrate limited overfitting through cross-validation.
- Produce an interpretable Decision Tree suitable for educational analysis.

DATASET PROFILING & TECHNICAL DESCRIPTION

The classification pipeline was developed using the Breast Cancer Wisconsin (Diagnostic) dataset available through the scikit-learn library.

Observation Unit

Each observation represents measurements extracted from a fine-needle aspirate (FNA) image of a breast mass.

Dataset Scale

- Total observations: 569
- Training observations: 455
- Testing observations: 114

- Numerical predictors: 30
- Missing values: 0
- Duplicate rows: 0

TARGET VARIABLE

Target

Breast tumor diagnosis

Positive Class

Malignant

Negative Class

Benign

For this study, the dataset labels were remapped so that:

- 1 = Malignant
- 0 = Benign

This allows evaluation metrics such as Sensitivity and Specificity to directly reflect disease detection performance.

PREDICTOR VARIABLES

The dataset contains 30 numerical features describing geometric and textural characteristics of breast cell nuclei, including:

- Mean Radius
- Mean Texture
- Mean Perimeter
- Mean Area
- Mean Smoothness
- Mean Compactness
- Mean Concavity

- Mean Concave Points
- Worst Radius
- Worst Perimeter
- Worst Concave Points

Feature importance analysis indicated that Worst Perimeter and Worst Concave Points contributed most strongly to the final Decision Tree.

PREPROCESSING & LEAKAGE CONTROL STRATEGY

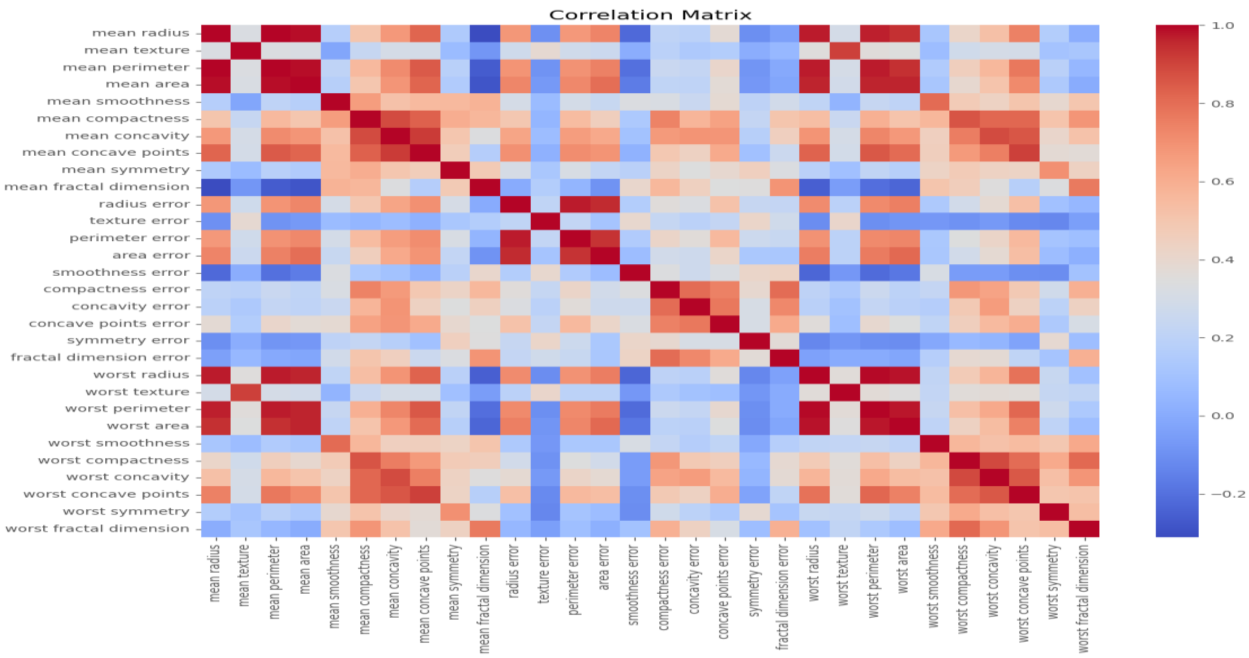
To ensure reproducibility and prevent information leakage, the dataset was divided into training and testing partitions before model development.

1. Dataset Partitioning

The dataset was split using:

- 80% Training
- 20% Testing
- Stratified sampling
- Random State = 42

Stratification preserved the malignant-to-benign class distribution across both partitions.

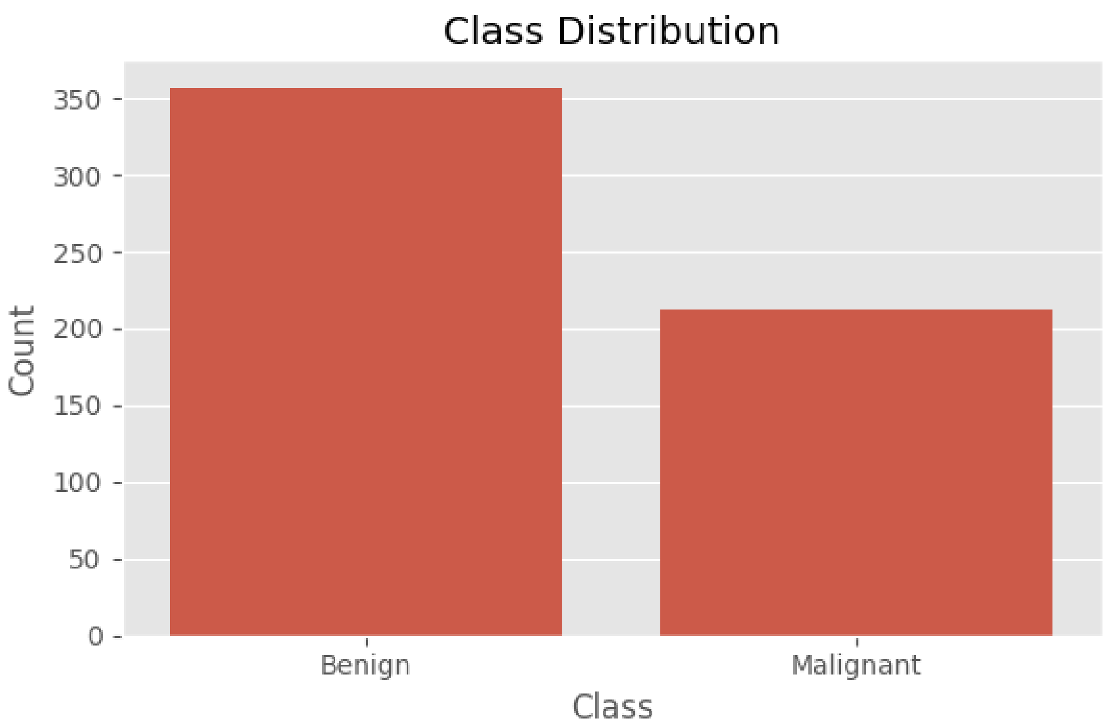


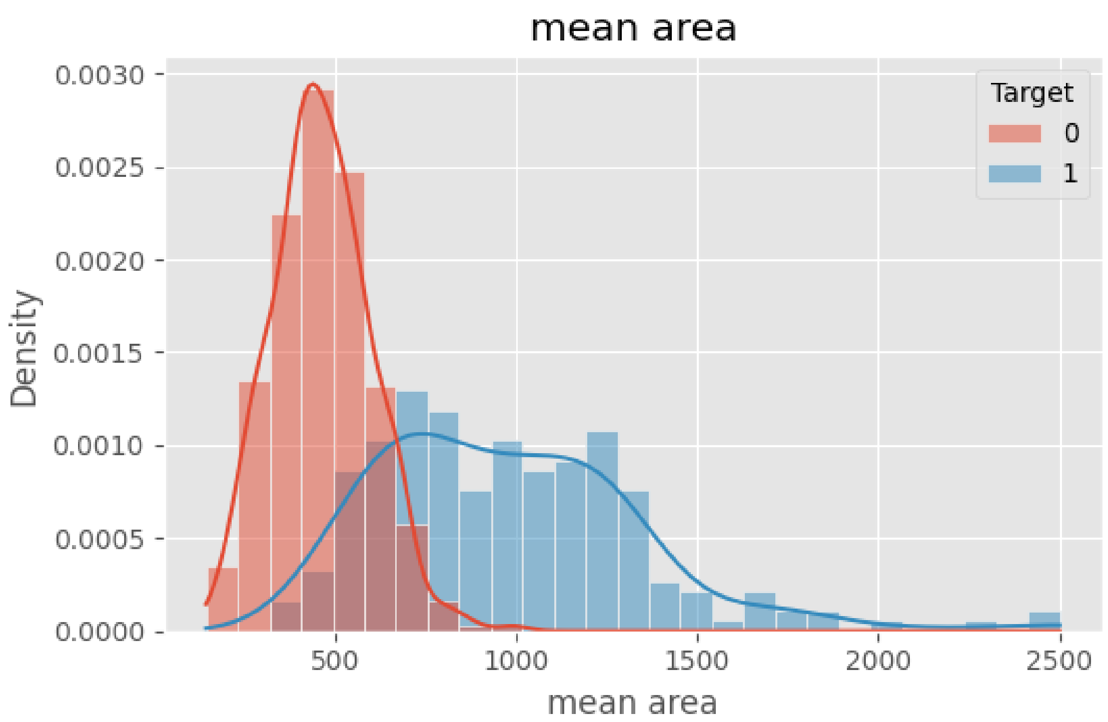
2. Data Audit

The dataset was inspected for:

- Missing values
- Duplicate observations
- Constant features
- Feature data types
- Class distribution

No missing values or duplicate rows were identified.





Dataset summary:

Dataset	rows	Unique patients	predictors	+ve class	+ve %	Missing %	Split method
Breast Cancer Wisconsin (Diagnostic)	569	569	30	Malignant	37.26%	0.00%	80–20 Stratified

3. Cross-Validation

Model selection was performed using 5-fold Stratified Cross-Validation on the training dataset.

This ensured that:

- Every training sample participated in validation exactly once.
- Model selection remained independent of the final test set.
- Generalization performance could be estimated before final evaluation.

- Cross-Validation Model Comparison

Model	ROC-AUC mean±SD	PR-AUC mean±SD	Sensitivity mean±SD	Specificity mean±SD	Balanced Acc. mean±SD	Complexity
Dummy	0.500 ± 0.000	0.374 ± 0.000	0.000 ± 0.000	—	0.500 ± 0.000	Baseline
Decision Tree	0.928 ± 0.035	0.862 ± 0.060	0.912 ± 0.059	—	0.928 ± 0.035	Depth = 8
Pruned Tree	0.949 ± 0.038	0.917 ± 0.071	0.871 ± 0.053	—	0.918 ± 0.036	Depth = 4

4. Model Complexity Control

Two Decision Tree configurations were evaluated:

- Unconstrained Decision Tree
- Pre-pruned Decision Tree (max_depth = 4, min_samples_leaf = 5)

The pruned tree reduced complexity from:

- Depth: 8 → 4
- Leaves: 24 → 9
- Nodes: 47 → 17

while maintaining similar predictive performance.

Tree Version	Criterion	Depth	Leaves	Nodes	ccp_alpha	CV Score	Interpretability Note
Original	Gini	8	24	47	0	0.932	Complex, higher risk of overfitting
Pruned	Gini	4	9	17	0	0.930	Simpler, easier to interpret

5. Leakage Prevention

To avoid optimistic performance estimates:

- The test set remained untouched during model development.
- Cross-validation was performed only on the training data.
- Hyperparameter choices were finalized before evaluating the test set.
- Final performance metrics were computed only once on the held-out test set.

Case Category	Observed Pattern	Possible Technical Explanation	Required Caution
False Negative	8 malignant cases predicted as benign	Borderline feature values or insufficient tree complexity	Do not infer medical cause from the model path.
False Positive	3 benign cases predicted as malignant	Feature overlap between benign and malignant samples	Consider prevalence and follow-up burden.
Unstable Prediction	Original tree deeper (Depth = 8); pruning reduced complexity (Depth = 4)	Full tree showed signs of overfitting; pruning improved interpretability	Check small leaves and split variability.
Out-of-Distribution Case	Model evaluated only on the Breast Cancer Wisconsin dataset	Performance on unseen hospitals/populations is unknown	Do not extrapolate beyond the training distribution.

FINAL TEST EVALS:

Accuracy: 0.9035087719298246

Balanced Accuracy: 0.8839285714285714

Precision: 0.918918918918919

Sensitivity (Recall): 0.8095238095238095

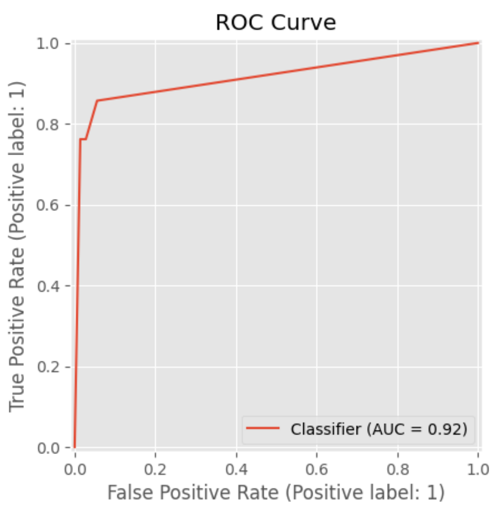
Specificity: 0.9583333333333334

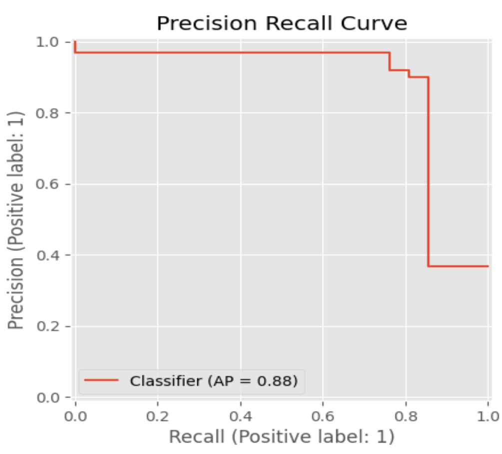
F1 Score: 0.8607594936708861

ROC AUC: 0.9153439153439153

PR AUC: 0.8780635043792939

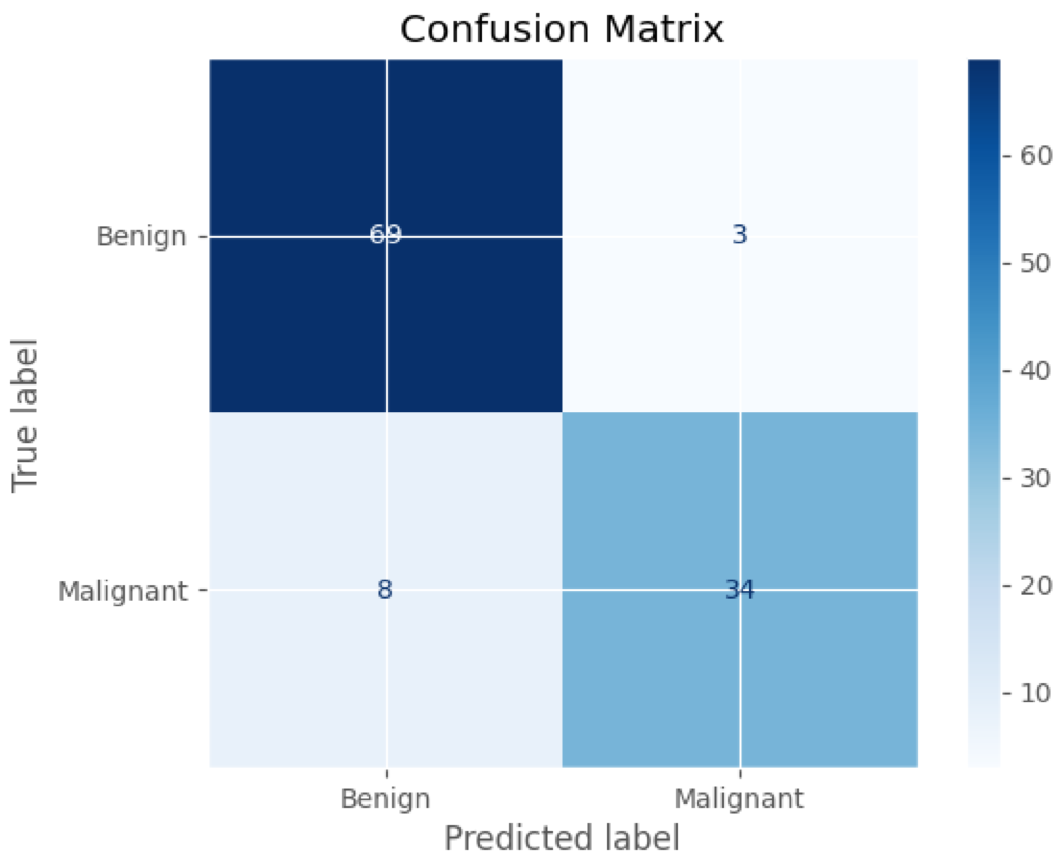
Brier Score: 0.0792293759210131

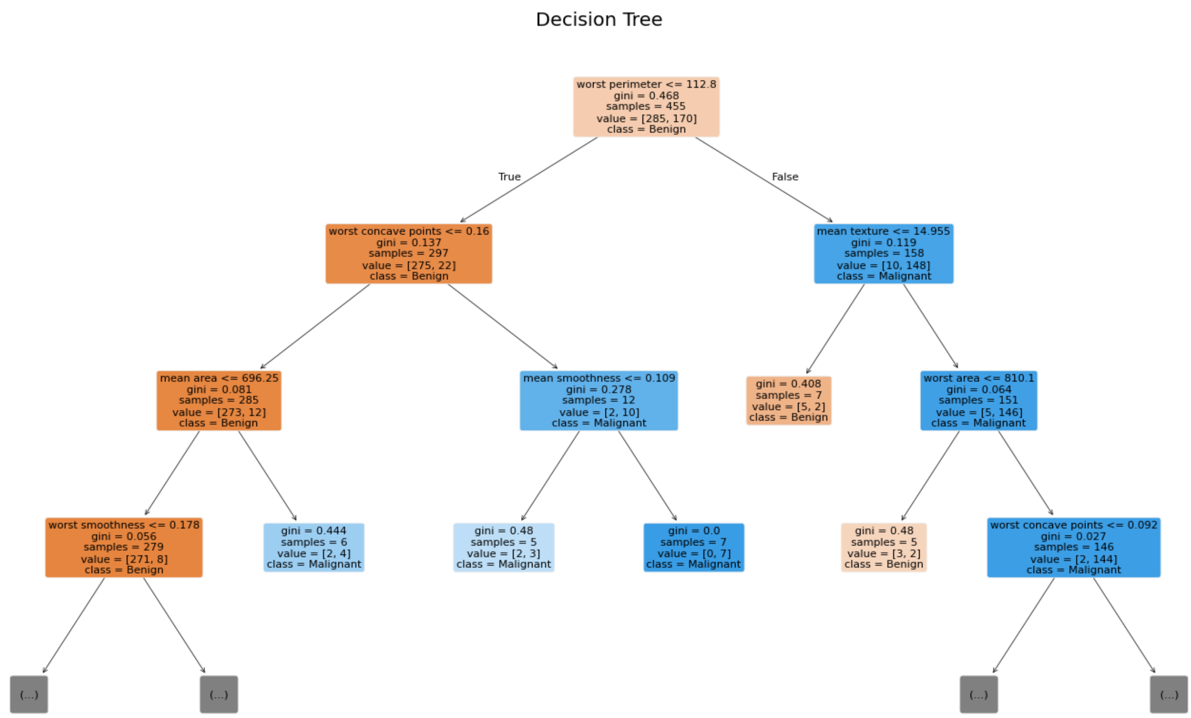




Classification Report

	precision	recall	f1-score	support
0	0.90	0.96	0.93	72
1	0.92	0.81	0.86	42
accuracy			0.90	114
macro avg	0.91	0.88	0.89	114
weighted avg	0.90	0.90	0.90	114





DISCUSSION:

The experimental study demonstrates that Decision Tree-based machine learning algorithms can effectively support disease classification using structured clinical measurements from the Breast Cancer Wisconsin dataset. A Dummy Classifier was first used as a baseline to establish the minimum expected predictive performance. Both the unconstrained Decision Tree and the pre-pruned Decision Tree substantially outperformed this baseline across all evaluation metrics.

On the final locked test set, the pre-pruned Decision Tree achieved an Accuracy of 90.35%, Sensitivity of 80.95%, Specificity of 95.83%, Precision of 91.89%, F1-Score of 86.08%, and ROC-AUC of 0.915. These results indicate that the model successfully distinguishes malignant and benign tumors while remaining simple enough for educational interpretation.

LIMITATIONS:

Despite achieving strong predictive performance, the developed classification system has several limitations:

- The dataset contains only 569 observations, which limits the diversity of training examples.
- Only structured numerical features derived from fine-needle aspirate measurements were used.
- Raw medical image analysis was outside the scope of this study.
- External validation using independent hospital datasets was not performed.
- The model was evaluated on a benchmark dataset and may not generalize to different clinical populations.
- The developed classifier is intended solely for educational and research purposes and must not replace professional medical diagnosis or clinical decision-making.

CONCLUSION:

This project successfully developed an interpretable disease classification system using Decision Tree-based machine learning techniques on the Breast Cancer Wisconsin (Diagnostic) dataset. Three stages of model development were evaluated: a Dummy Classifier baseline, an unconstrained Decision Tree, and a pre-pruned Decision Tree.

Experimental results demonstrated that the Decision Tree models substantially outperformed the baseline classifier. Although the unconstrained Decision Tree achieved excellent training performance, cross-validation indicated evidence of overfitting. Applying pre-pruning reduced tree complexity while maintaining strong predictive performance and improving model interpretability.

The final pre-pruned Decision Tree achieved an Accuracy of 90.35%, ROC-AUC of 0.915, and Specificity of 95.83%, demonstrating that interpretable machine learning models can provide reliable predictions for disease classification within the limitations of the benchmark dataset.

Overall, this study illustrates the value of Decision Trees as transparent decision-support models for healthcare analytics and demonstrates the importance of balancing predictive performance with interpretability and responsible evaluation.

REFERENCES:

1. UCI Machine Learning Repository. *Breast Cancer Wisconsin (Diagnostic) Dataset*.
2. L. Breiman. *Random Forests*. Machine Learning, 2001.
3. J. R. Quinlan. *Induction of Decision Trees*. Machine Learning, 1986.
4. Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.
5. Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning*. Springer, 2009.

Github: https://github.com/Thilipan55/Advanced_pred_analysis_Thilipan-2.git